

Higher-Order Quantum Operations

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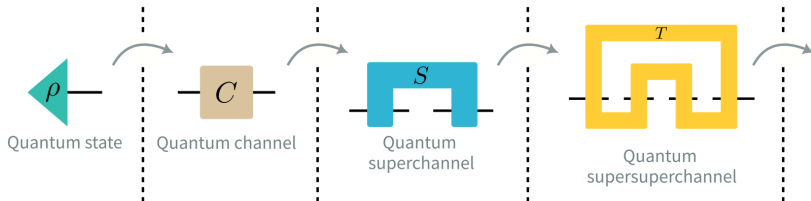
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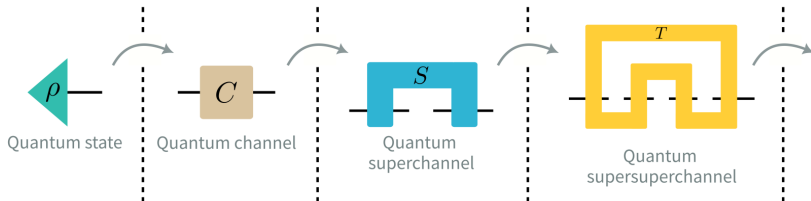


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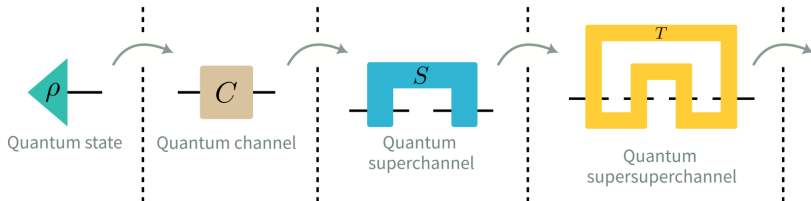


What are higher-order quantum operations?



Why, how, and when to use higher-order operations?

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Higher-Order Quantum Operations

Philip Taranto, Simon Milz, Mio Murao, Marco Túlio Quintino, Kavan Modi

Why, how, and when to use higher-order operations?

Presentation outline

- ▶ Warming up: Transforming quantum states

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- ▶ Part 1: Transforming quantum operations

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- ▶ Part 2: Measuring quantum operations

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- ▶ Part 3: Transformations beyond the circuit formalism

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- ▶ Part 1: Transforming quantum operations
- ▶ Part 2: Measuring quantum operations
- ▶ Part 3: Transformations beyond the circuit formalism
- ▶ Part 4: The cost of a quantum circuit simulation

Warming up:

How to transform quantum states?

Quantum state

$$\rho \in \mathcal{L}(\mathcal{H}) \cong \mathcal{L}(\mathbb{C}^d)$$

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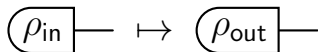
$$\boxed{\rho} \text{---} \in \mathcal{L}(\mathcal{H})$$

How to transform quantum states?

$$\rho_{\text{in}} \mapsto \rho_{\text{out}}$$

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Maths oriented quantum info

- ▶ Linear maps:

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$$\boxed{\rho_{\text{in}}} \text{ --- } \in \text{state} \quad \Longrightarrow \quad \boxed{\rho_{\text{in}}} \text{ --- } \boxed{\tilde{C}} \text{ --- } \in \text{state}$$

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- ▶ OK... but how about state transposition, $\tilde{T}(\rho) = \rho^T$?

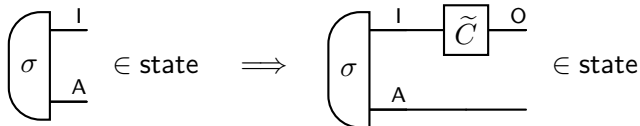
Physics oriented quantum info

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 is a quantum channel when



Physics oriented quantum info

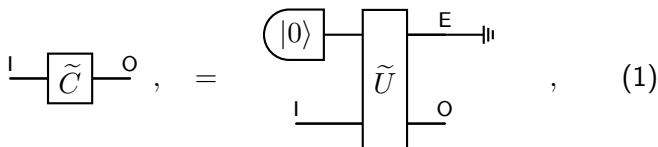
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Physics oriented quantum info

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The diagram shows an equation between two quantum circuit representations. On the left, a horizontal line labeled 'I' enters a square box labeled $\tilde{C}$, and a horizontal line labeled 'O' exits the box. This is followed by a comma and an equals sign. On the right, a vertical rectangle labeled $\tilde{U}$ has two inputs: a horizontal line labeled 'I' at the bottom and a vertical line labeled $|0\rangle$ at the top. It has two outputs: a horizontal line labeled 'O' at the bottom and a vertical line labeled 'E' at the top. The 'E' output line ends with a double vertical bar symbol, representing a trace operation. This is followed by a comma and the label (1).

$$I \rightarrow \boxed{\tilde{C}} \rightarrow O, \quad = \quad \begin{array}{c} |0\rangle \\ \downarrow \\ \boxed{\tilde{U}} \\ \uparrow \\ I \end{array} \begin{array}{l} \text{E} \\ \text{O} \end{array}, \quad (1)$$

- ▶ Nice, this is CPTP!

Physics oriented quantum info

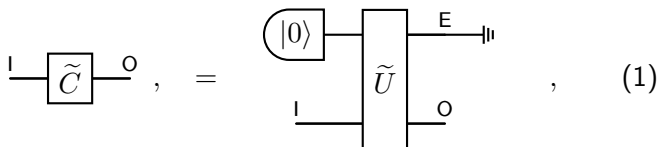
- ▶ Schrodinger equation, unitary operations
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$$\text{I} \xrightarrow{\tilde{C}} \text{O} , \quad = \quad \begin{array}{c} |0\rangle \\ \text{I} \end{array} \xrightarrow{\tilde{U}} \begin{array}{c} \text{E} \\ \text{O} \end{array} , \quad (1)$$

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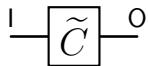
- ▶ Nice, this is CPTP!
- ▶ Naimark dilation: CPTP is this!
- ▶ We're all happy! Pick your favourite approach. :)

Part 1

Trasnforming quantum channels

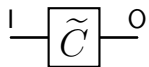
Quantum channels

$$\tilde{C} : \mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O)$$



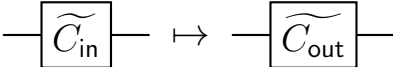
Quantum channels

$$\tilde{C} : \mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O)$$



$\tilde{C}$ is CPTP

How to transform quantum channels?

$$\widetilde{C}_{\text{in}} \mapsto \widetilde{C}_{\text{out}}$$


The diagram illustrates the transformation of a quantum channel. It consists of two parts connected by a mapping symbol $\mapsto$. On the left, a horizontal line enters a square box labeled $\widetilde{C}_{\text{in}}$, and another horizontal line exits the box to the right. On the right, a horizontal line enters a square box labeled $\widetilde{C}_{\text{out}}$, and another horizontal line exits the box to the right.

Definitely, this can be done

- Pre-processing and post-processing:

$$\tilde{\tilde{S}}(\tilde{C}) = \text{I}' \text{---} \boxed{\tilde{E}} \text{---} \text{I} \text{---} \boxed{\tilde{C}} \text{---} \text{O} \text{---} \boxed{\tilde{D}} \text{---} \text{O}'$$

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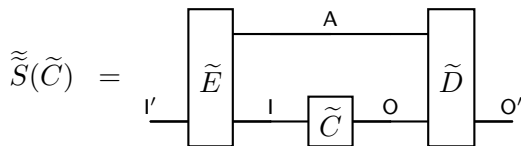
$$\tilde{\tilde{S}}(\tilde{C}) = \text{---}' \boxed{\tilde{E}} \text{---}^{\text{I}} \boxed{\tilde{C}} \text{---}^{\text{O}} \boxed{\tilde{D}} \text{---}^{\text{O}'}$$

- As a supermap $\tilde{\tilde{S}} : [\mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O)] \rightarrow [\mathcal{L}(\mathcal{H}_{I'}) \rightarrow \mathcal{L}(\mathcal{H}_{O'})]$

$$\tilde{\tilde{S}}(\tilde{C}) = \tilde{D} \circ \tilde{C} \circ \tilde{E}$$

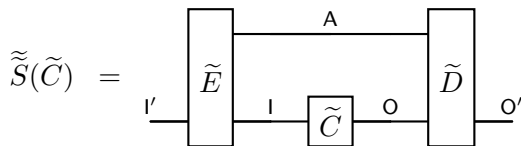
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$$\tilde{S}(\tilde{C}) = \text{tr}_A \left(\tilde{D} \circ \left(\tilde{C} \otimes \tilde{I}_A \right) \circ \tilde{E} \right)$$

What can we accept?

- Quantum superchannel:

$$\widetilde{\widetilde{S}} : [\mathcal{L}(\mathcal{H}_I) \rightarrow \mathcal{L}(\mathcal{H}_O)] \rightarrow [\mathcal{L}(\mathcal{H}_{I'}) \rightarrow \mathcal{L}(\mathcal{H}_{O'})]$$

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if $\widetilde{C}$ is TP, then $\widetilde{\widetilde{S}}(\widetilde{C})$ is TP

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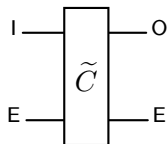
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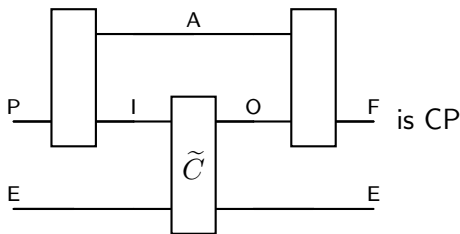
- ▶ CP-Preserving

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- ▶ Well.. Completely CP-preserving



is CP, then



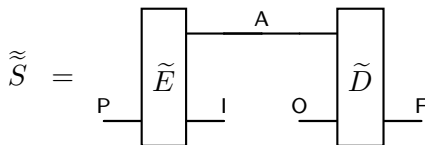
is CP

Quantum superchannels

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Quantum superchannels

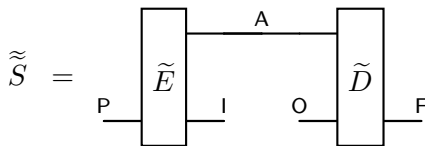
- ▶ We are all happy! :)
- ▶ $\tilde{\tilde{S}}$ is TPP and CCPP iff $\exists$ quantum channels $\tilde{E}$ and $\tilde{D}$ such that $\forall \tilde{C}$, we have $\tilde{\tilde{S}}(\tilde{C}) = \text{tr}_A \left(\tilde{D} \circ \left(\tilde{C} \otimes \tilde{I}_A \right) \circ \tilde{E} \right)$, that is,



G. Chiribella, G. M. D'Ariano, and P. Perinotti EPL (2008)

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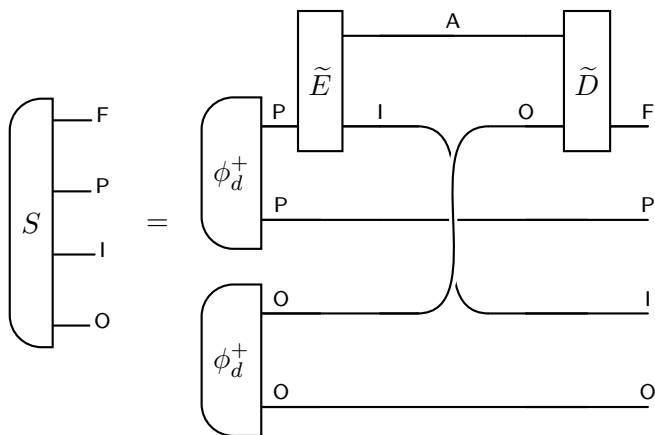


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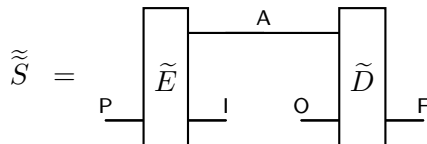
- ▶ Causality is proven!

Characterising quantum superchannels

Choi isomorphism for supermaps:



Characterizing quantum superchannels



$S \in \mathcal{L}(\mathcal{H}_P \otimes \mathcal{H}_I \otimes \mathcal{H}_O \otimes H_F)$ is a superchannel iff

$$S \geq 0$$

$$\text{tr}_F(S) = \text{tr}_{OF}(S) \otimes \frac{I_F}{d_F}$$

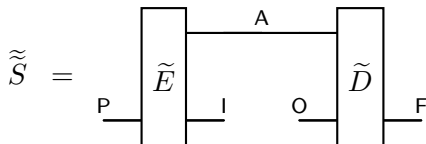
$$\text{tr}_{IOF}(S) = \text{tr}_{PIOF}(S) \otimes \frac{I_I}{d_I}$$

$$\text{tr}(S) = d_P d_O$$

G. Chiribella, G.M D'Ariano, P. Perinotti, PRL (2008)

G. Gutoski, J. Watrous, STOC (2007)

Characterizing quantum superchannels



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Affine and positive semidefinite constraints $\implies$ SDP!!

Quantum unitary transformations

$$\text{---} \boxed{\widetilde{U}} \text{---} \mapsto \text{---} \boxed{\widetilde{f(U)}} \text{---}$$

What do we want?

Ideally...

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Ideally...

Something like this:

$$\text{---} \boxed{\widetilde{Y}} \text{---} \boxed{\widetilde{U}_2} \text{---} \boxed{\widetilde{Y}} \text{---} = \text{---} \boxed{\widetilde{U}_2^*} \text{---}$$

Phys. Rev. Research (2019)

J. Miyazaki, A. Soeda, and M. Murao

Applications

- ▶ Unitary conjugation, $f(U) = U^*$:

$$F(d) = \frac{2}{d(d-1)}, \text{ or } p(d > 2) = 0.$$

G. Chiribella, D. Ebler, NJP (2016), J. Miyazaki, A. Soeda, M. Murao, PRR (2019)

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- ▶ Unitary transposition, $f(U) = U^T$:

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M.T. Quintino, D. Ebler, Quantum (2022), M.T. Quintino, Q. Dong, A. Shimbo, A. Soeda, M. Murao, PRA (2019)

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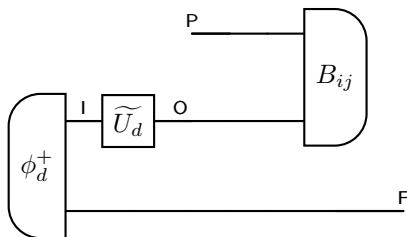
G. Chiribella, D. Ebler, NJP (2016), J. Miyazaki, A. Soeda, M. Murao, PRR (2019)

- ▶ Unitary Storage-and-Retrieval $f(U) = U$:

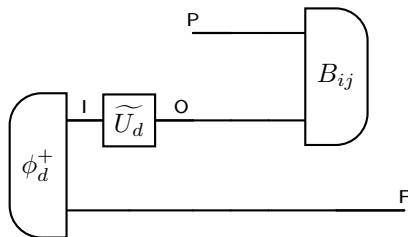
A. Bisio, G. Chiribella, G. M. D'Ariano, S. Facchini, P. Perinotti M. Sedlák, PRA (2010) A.

Bisio, M. Ziman, PRL (2019)

Explicit construction for unitary transposition

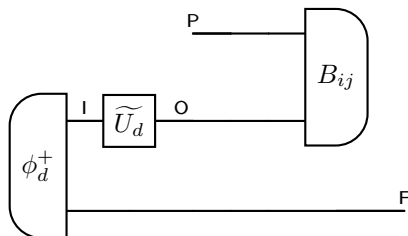


Explicit construction for unitary transposition



$$(U \otimes \mathbf{1}) |\phi_d^+\rangle = (\mathbf{1} \otimes U^T) |\phi_d^+\rangle$$

Explicit construction for unitary transposition

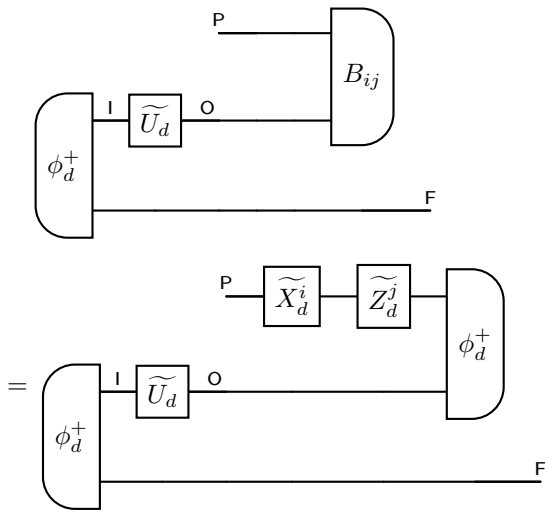


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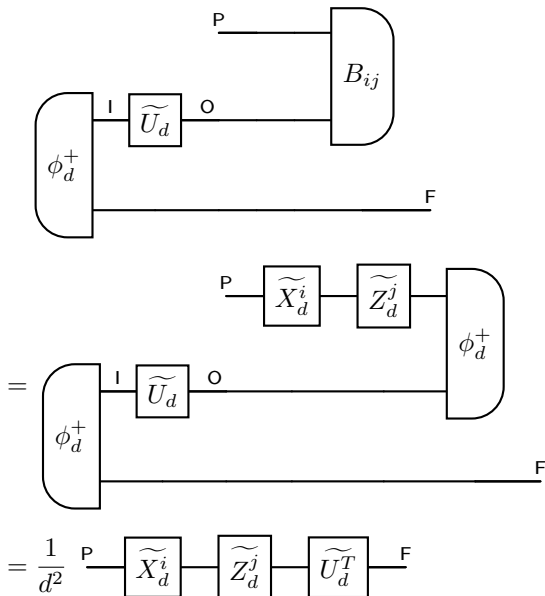
$$B_{ij} := \left(Z_d^j X_d^i \otimes \mathbf{1} \right) |\phi_d^+\rangle$$

$$i, j \in \{0, \dots, d-1\}$$

Explicit construction for unitary transposition



Explicit construction for unitary transposition



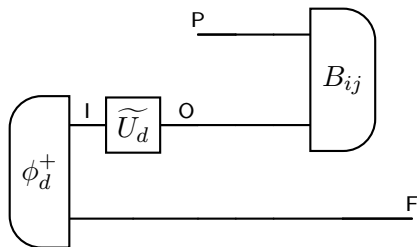
Qubit unitary inversion

For qubits, we have:

$$\text{---} \boxed{\widetilde{Y}} \text{---} \boxed{\widetilde{U_2}} \text{---} \boxed{\widetilde{Y}} \text{---} = \text{---} \boxed{\widetilde{U_2^*}} \text{---}$$

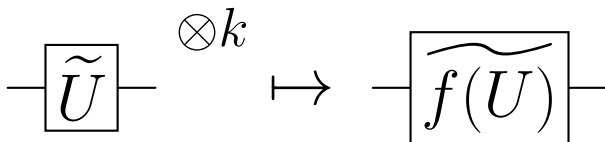
Hence, with $p = 1/4$, we can invert an arbitrary unitary operation!

Delayed input state



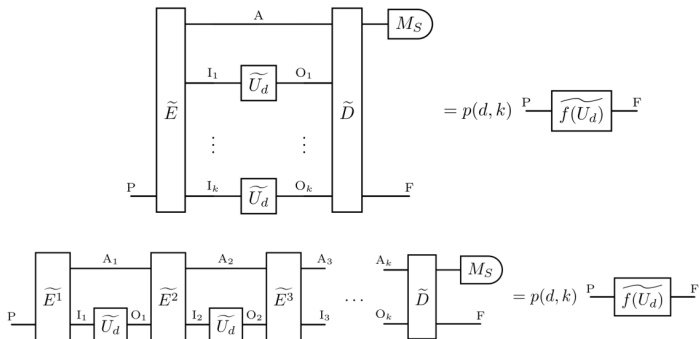
Quantum unitary transformations

We can call/query operations many times!



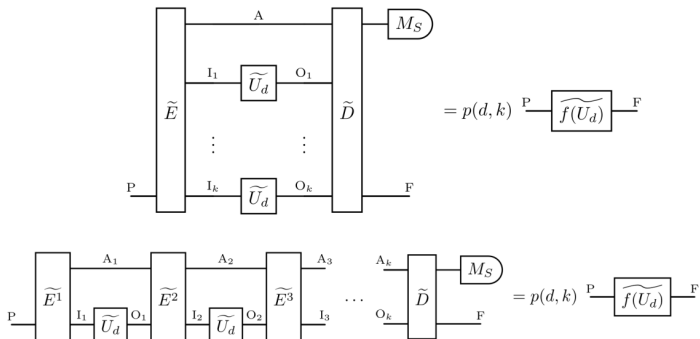
Multiple calls

► Scenarios with multiple calls:



Multiple calls

- Scenarios with multiple calls:



- This also fits the higher-order quantum operations framework

The homomorphic case

- ▶ When $f(UV) = f(U)f(V)$, parallel calls are optimal
A. Bisio, G. M. D'Ariano, P. Perinotti, M. Sedlak, PLA (2014)

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A. Bisio, G. M. D'Ariano, P. Perinotti, M. Sedlak, PLA (2014)
- ▶ If $f : \text{SU}(d) \rightarrow \text{SU}(d)$, $f(U) = U^*$ is the only non-trivial homomorphism

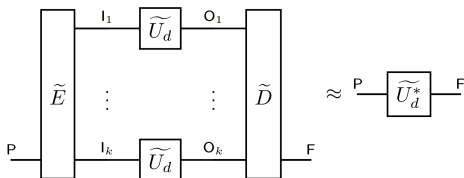
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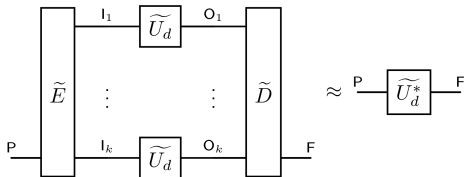
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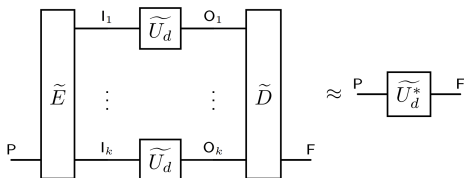
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IEEE Trans. Inf. Theory (2022)

D. Ebler, M. Horodecki, M. Marciniak, T. Młynik, M.T. Quintino, M. Studziński

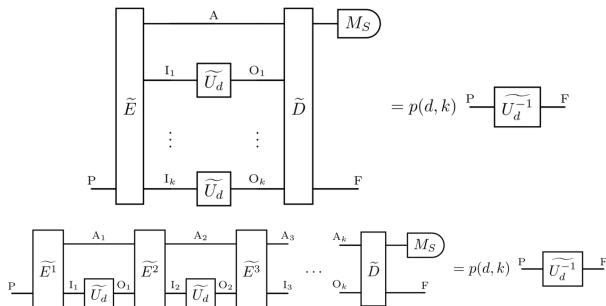
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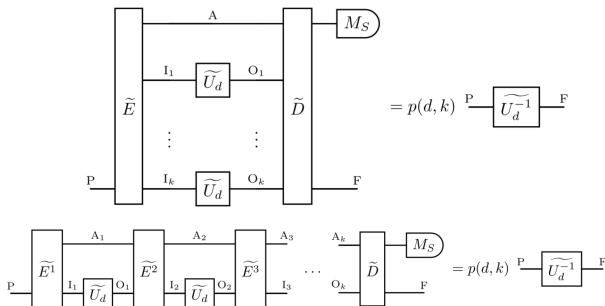
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IEEE Trans. Inf. Theory (2022)
D. Ebler, M. Horodecki, M. Marciniak, T. Młynik, M.T. Quintino, M. Studziński
- ▶ Also, if $k < d - 1$, $p = 0$
M.T. Quintino, Q. Dong, A. Shimbo, A. Soeda, M. Murao
PRA (2019)

Unitary inversion



► Parallel ($d = 2$): $p = 1 - \frac{3}{k+3}$

Unitary inversion

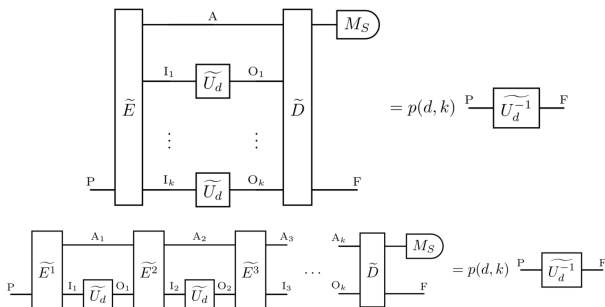


► Parallel ($d = 2$): $p = 1 - \frac{3}{k+3}$

► Parallel ($k \geq d - 1$):

$$1 - \frac{1}{k} \sim 1 - \frac{d^2 - 1}{\lfloor \frac{k}{d-1} \rfloor + d^2 - 1} \leq p \leq 1 - \frac{d^2 - 1}{k(d-1) + d^2 - 1} \sim 1 - \frac{1}{k}$$

Unitary inversion



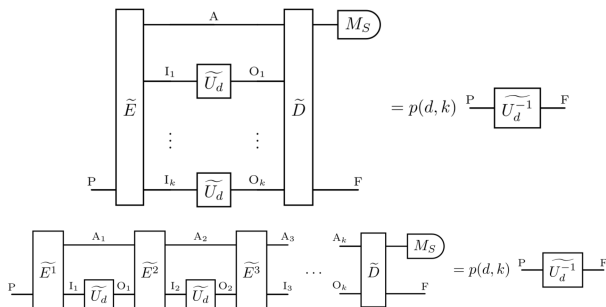
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► Optimal parallel $\implies$ delayed input-state

Unitary inversion



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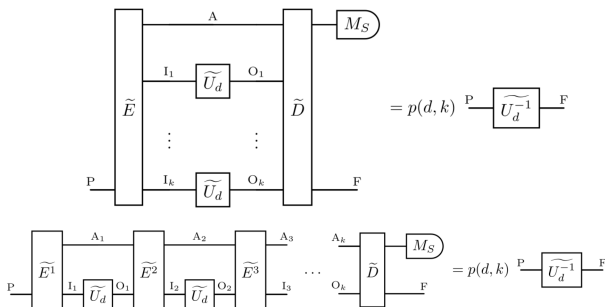
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► If $k < d - 1$, then $p = 0$

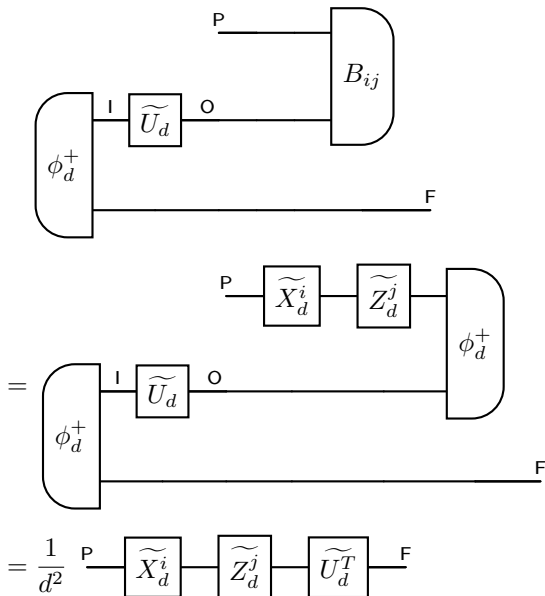
Unitary inversion



- ▶ Parallel ($d = 2$): $p = 1 - \frac{3}{k+3}$
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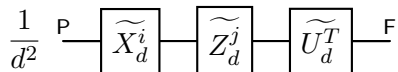
$$1 - \frac{1}{k} \sim 1 - \frac{d^2 - 1}{\lfloor \frac{k}{d-1} \rfloor + d^2 - 1} \leq p \leq 1 - \frac{d^2 - 1}{k(d-1) + d^2 - 1} \sim 1 - \frac{1}{k}$$
- ▶ Optimal parallel $\implies$ delayed input-state
- ▶ If $k < d - 1$, then $p = 0$
- ▶ Sequential ($k \geq d - 1$): $p \geq 1 - \left(1 - \frac{1}{d^2}\right)^{\lceil \frac{k+2-d}{d} \rceil} \sim 1 - \frac{1}{e^k}$

Unitary transposition



Adaptive circuits

When we fail, we lose track of the unknown input state...
then we cannot re-iterate this protocol...



Adaptive circuits

Can we “reset the protocol” when we fail?

$$\text{P} \text{ --- } \boxed{\widetilde{X}_d^i} \text{ --- } \boxed{\widetilde{Z}_d^j} \text{ --- } \boxed{\widetilde{U}_d^T} \text{ --- } \boxed{\widetilde{C}} \text{ --- } \text{F} = \text{P} \text{ --- } \boxed{\widetilde{\mathbb{1}}_d} \text{ --- } \text{F}$$

Adaptive circuits

Sometimes we can reset it.

$$P \text{ --- } \boxed{\widetilde{X}_d^i} \text{ --- } \boxed{\widetilde{Z}_d^j} \text{ --- } \boxed{\widetilde{U}_d^T} \text{ --- } \boxed{\widetilde{U}_d^*} \text{ --- } \boxed{\widetilde{Z}_d^{-j}} \text{ --- } \boxed{\widetilde{X}_d^{-i}} \text{ --- } F = P \text{ --- } \boxed{\widetilde{\mathbb{1}}_d} \text{ --- } F$$

Success or draw $\implies$ repeat until success (approaches one exponentially)

Arbitrary functions $f(U_d)$

Nice!

Success or draw strategy exists for inverse and transposition!

Arbitrary functions $f(U_d)$

Nice!

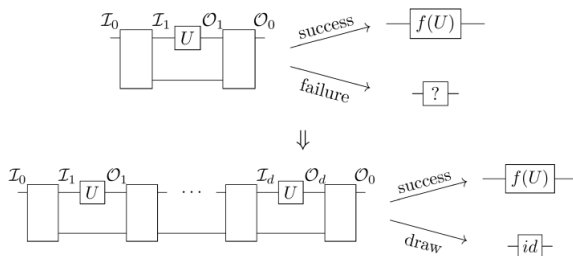
Success or draw strategy exists for inverse and transposition!
but how about other functions of unitaries?

$$U_d^{\otimes k} \mapsto f(U_d)$$

Success or draw

Theorem

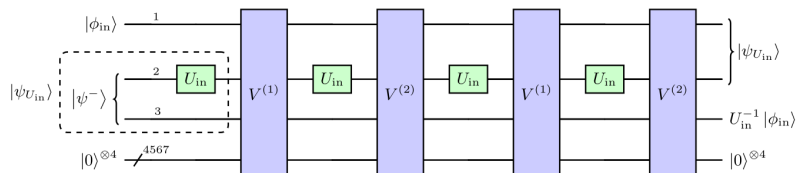
With $k = d$ calls, success or draw is always possible!



Q. Dong, M.T. Quintino, A. Soeda, M. Murao
PRL (2021)

Deterministic and exact is possible

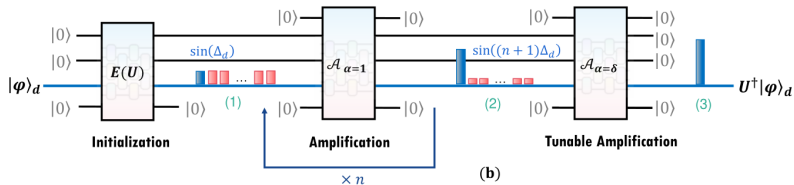
For qubits, $k = 4$ calls are enough:



S. Yoshida, A. Soeda, M. Murao PRL (2023)

Success or draw

And, for qudits, $k \approx d^2$,



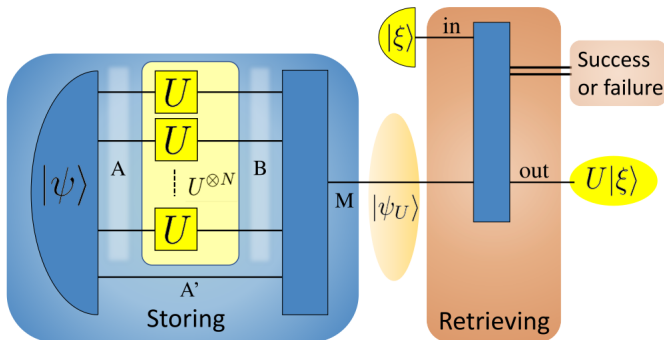
Y. A. Chen, Y. Mo, Y. Liu, L. Zhang, X. Wang (2024)

Probabilistic parallel unitary transposition

Great, and what can we say about parallel probabilistic unitary transposition/inversion?

Probabilistic parallel unitary transposition

Equivalent to unitary SAR¹, can be done with probabilistic PBT

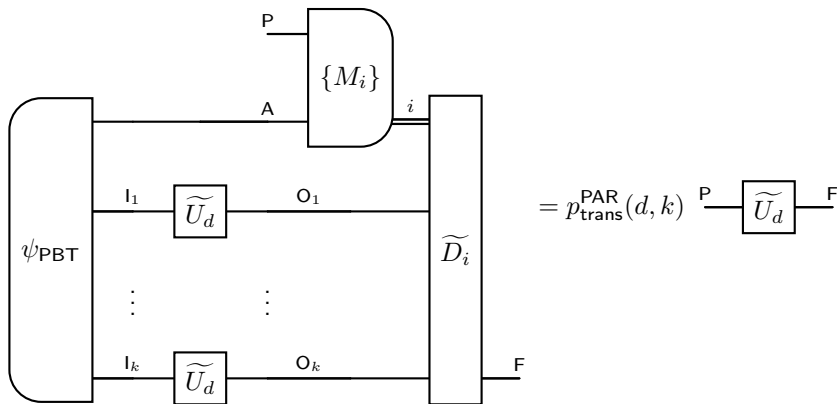


¹ : Optimal Probabilistic Storage and Retrieval of Unitary Channels
M. Sedláč, A. Bisio, and M. Ziman
PRL 2019

Probabilistic parallel unitary transposition

Equivalent to unitary SAR, can be done with probabilistic PBT

$$p_{\text{trans}}^{\text{PAR}}(d, k) = 1 - \frac{d^2 - 1}{k + d^2 - 1}, \quad (U_d \otimes \mathbb{1})^{\otimes k} |\psi_{\text{PBT}}\rangle = (\mathbb{1} \otimes U_d^T)^{\otimes k} |\psi_{\text{PBT}}\rangle$$

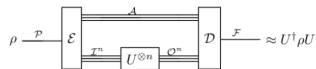


Deterministic parallel unitary inversion and transposition

What does it change in a deterministic non-exact scenario?

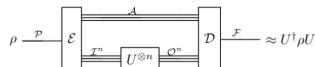
Deterministic parallel unitary inversion and transposition

(d) Parallel unitary inversion



Deterministic parallel unitary inversion and transposition

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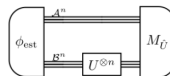


Thm. 4 $F_{\text{inv}}^{(\text{PAR})}(n, d) = F_{\text{inv}}^{(\text{SEQ})}(n, d)$
 $= F_{\text{inv}}^{(\text{GEN})}(n, d) = \frac{n+1}{d^2} \text{ for } n \leq d-1$

One-to-one Correspondence between Deterministic Port-Based Teleportation and Unitary Estimation
 S. Yoshida, Y. Koizumi, M. Studziński, M.T. Quintino, M. Murao
 arXiv:2408.11902 (2024)

Deterministic parallel unitary inversion and transposition

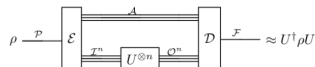
(b) Unitary estimation



$$F_{\text{est}}(n, d) = F_{\text{inv}}^{(\text{PAR})}(n, d)$$

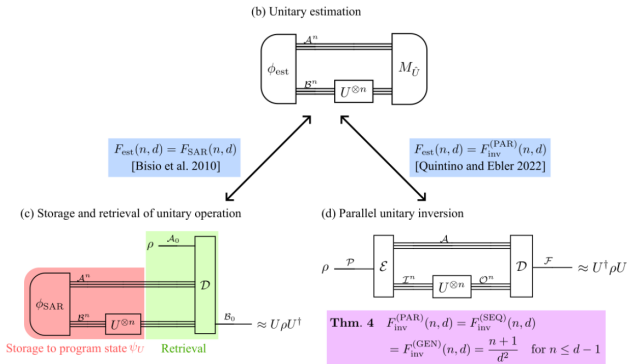
[Quintino and Ebler 2022]

(d) Parallel unitary inversion



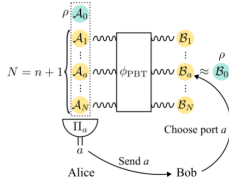
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Deterministic parallel unitary inversion and transposition

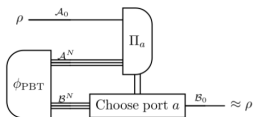


Deterministic parallel unitary inversion and transposition

(a-1) Deterministic port-based teleportation (dPBT)



(a-2) Quantum circuit for dPBT



$$F_{\text{PBT}}(n+1, d) = F_{\text{est}}(n, d)$$

This work (Thm. 1)

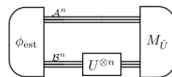
Cor. 2 $F_{\text{PBT}}(N, d) = 1 - \Theta(d^4 N^{-2})$

Cor. 3 $F_{\text{est}}(n, d) = \frac{n+1}{d^2}$ for $n \leq d-1$

$$F_{\text{est}}(n, d) = F_{\text{SAR}}(n, d)$$

[Bisio et al. 2010]

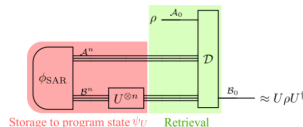
(b) Unitary estimation



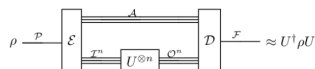
$$F_{\text{est}}(n, d) = F_{\text{inv}}^{(\text{PAR})}(n, d)$$

[Quintino and Ebler 2022]

(c) Storage and retrieval of unitary operation



(d) Parallel unitary inversion



Thm. 4 $F_{\text{inv}}^{(\text{PAR})}(n, d) = F_{\text{inv}}^{(\text{SEQ})}(n, d)$
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Part 2

Measuring quantum channels

Measuring quantum channels:

- ▶ Quantum measurement:

$$\rho \mapsto p_i$$

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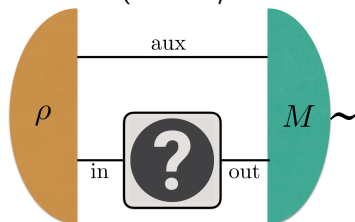
$$\tilde{C} \mapsto p_i$$

- ▶ PPOVM/testers : $T_i \geq 0, \sum_i M_i = \sigma \otimes \mathbb{1}, \text{tr}(\sigma) = 1$

$$\text{tr}(T_i C) = p_i$$

Measuring quantum operations

Super POVMs! (Testers/Process POVMs)



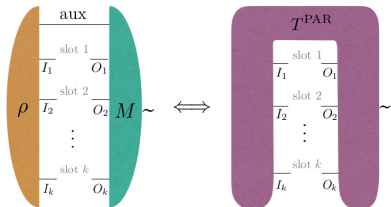
Process POVM: A mathematical framework for the description of process tomography experiments M. Ziman, PRA (2008)

Theoretical framework for quantum networks

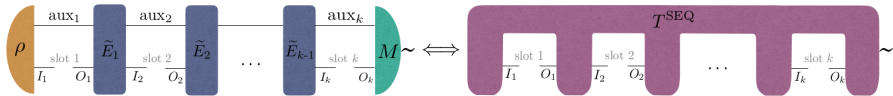
G. Chiribella, G.M. D'Ariano, P. Perinotti, PRA (2009)

Measuring quantum operations

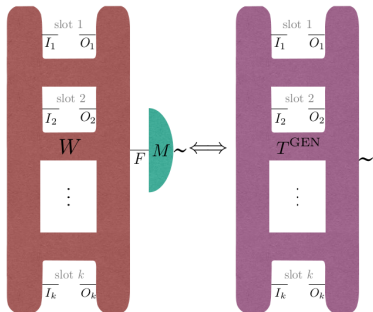
(a) PARALLEL



(b) SEQUENTIAL



(c) GENERAL



G. Chiribella, G.M. D'Ariano, P. Perinotti PRL (2008)

G. Chiribella, G.M. D'Ariano, P. Perinotti, PRA (2009)

J. Bavaresco, M. Murao, M.T. Quintino PRL (2021)

J. Bavaresco, M. Murao, M.T. Quintino J. Math. Phys. (2022)

Channel discrimination

INPUT: k calls/queries of 

PROMISE:  $\in \left\{ \begin{array}{c} \text{---} \text{  \text{---} \\ \tilde{C}_1 \\ p_1 \end{array} , \begin{array}{c} \text{---} \text{  \text{---} \\ \tilde{C}_2 \\ p_2 \end{array} , \begin{array}{c} \text{---} \text{  \text{---} \\ \tilde{C}_3 \\ p_3 \end{array} \right\}$

TASK: guess which channel  is.

Applications:

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Finding optimal strategies, when sequential strategies are useful

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A. Abbott, M. Mhalla, P. Pocreau, PPR (2024)

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- ▶ Quantum measurement discrimination: M. Sedlak, M. Ziman, PRA (2014)

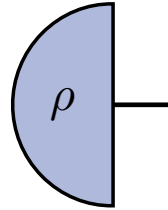
Part 3

Transformations beyond the circuit formalism

Formalism: Higher-order operations

quantum data:

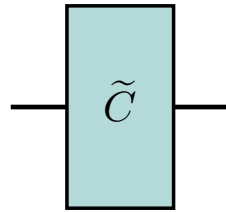
quantum states



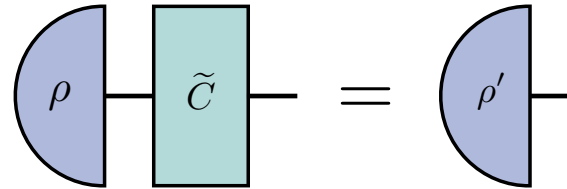
$$\rho \in \mathcal{L}(\mathcal{H}_{\text{in}})$$

quantum functions:

quantum operations
(quantum channels)



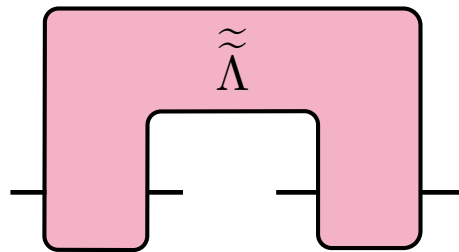
:



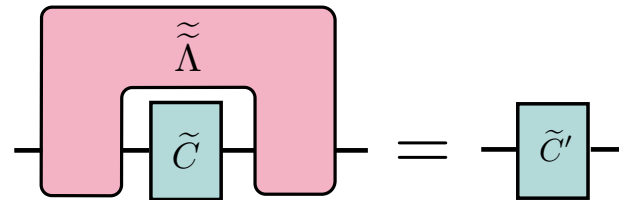
$$\tilde{C} : \mathcal{L}(\mathcal{H}_{\text{in}}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{out}})$$

higher-order quantum operations:

quantum processes



:



$$\begin{aligned} \tilde{\tilde{\Lambda}} &: [\mathcal{L}(\mathcal{H}_{\text{in}}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{out}})] \\ &\rightarrow [\mathcal{L}(\mathcal{H}_{\text{in}'}) \rightarrow \mathcal{L}(\mathcal{H}_{\text{out}'})] \end{aligned}$$

“functions of functions”

Key features: Higher-order operations

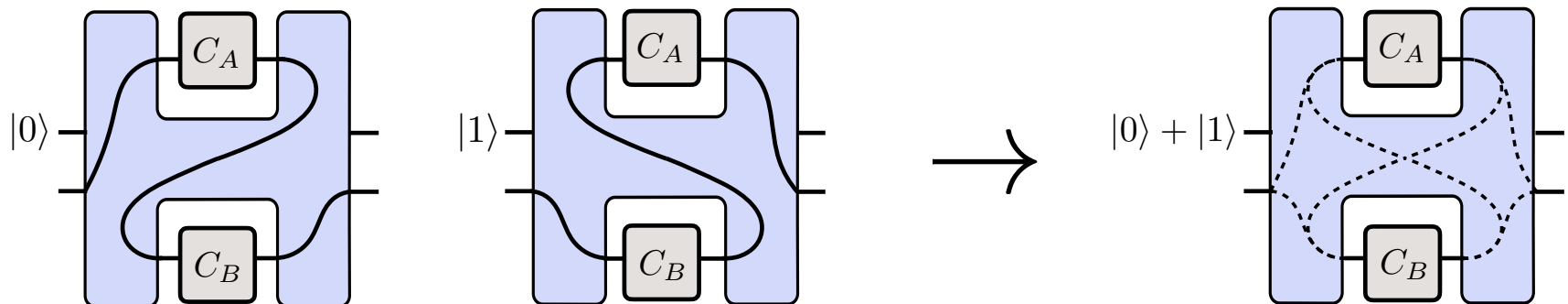
2) Quantum higher-order operations go beyond the quantum circuit model

CLASSICAL	QUANTUM	HIGHER-ORDER QUANTUM
data (bits) gates circuits	quantum data (qbits) quantum gates circuits	quantum data (qbits) quantum gates quantum circuit structure

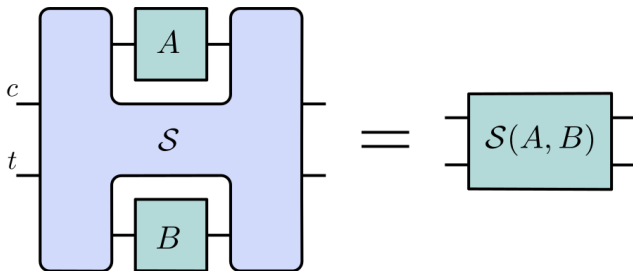
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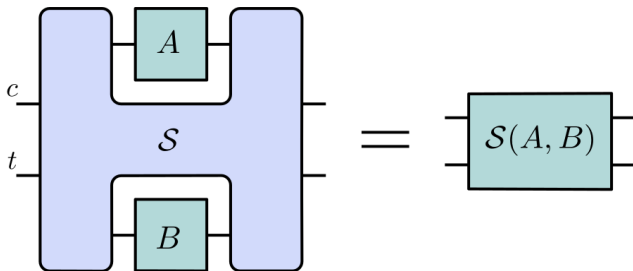


The quantum switch



G. Chiribella, G. M. D'Ariano, P. Perinotti, B. Valiron, PRA (2013)

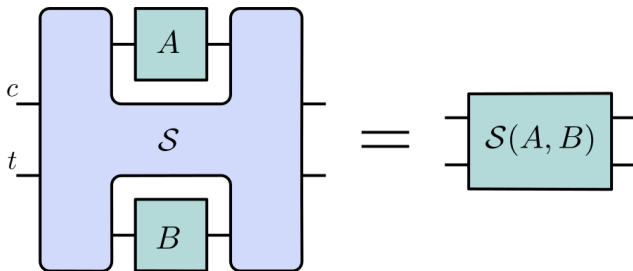
The quantum switch



G. Chiribella, G. M. D'Ariano, P. Perinotti, B. Valiron, PRA (2013)

If $A(\rho) = U_A \rho U_A^\dagger$ and $B(\rho) = U_B \rho U_B^\dagger$

The quantum switch



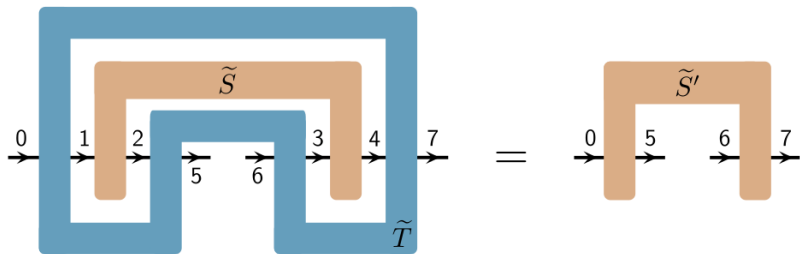
G. Chiribella, G. M. D'Ariano, P. Perinotti, B. Valiron, PRA (2013)

If $A(\rho) = U_A \rho U_A^\dagger$ and $B(\rho) = U_B \rho U_B^\dagger$

$$\mathcal{S} : (U_A, U_B) \mapsto |0\rangle\langle 0| \otimes U_B U_A + |1\rangle\langle 1| \otimes U_A U_B$$

Beyond the circuit formalism

The superper channel:

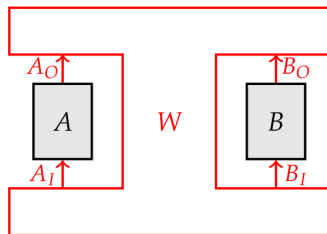


S. Milz, J. Bavaresco, G. Chiribella, Quantum (2022) S. Milz, M.T. Quintino, Quantum (2024)

Beyond the circuit formalism

Process matrices: extracting probabilities from quantum instruments:

$$\text{tr}(W A_{a|x} \otimes B_{b|y}) = p(ab|xy)$$



O. Oreshkov, F. Costa, C. Brukner, Nature Communications (2012)
M. Araújo, C. Branciard, F. Costa, A. Feix, C. Giarmatzi, C. Brukner, NJP (2015)

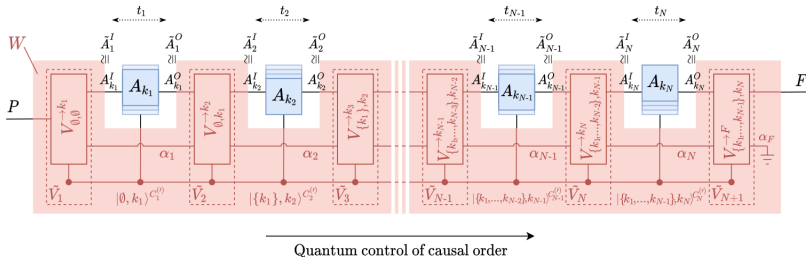
Beyond the circuit formalism

Is indefinite causality OK?

Beyond the circuit formalism

What can be done?

Quantum circuits with quantum control:

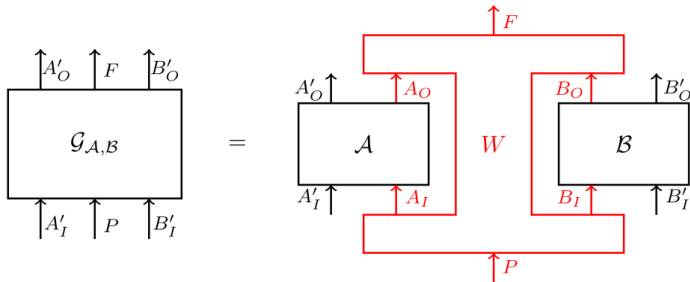


J. Wechs, H. Dourdent, A.A. Abbott, C. Branciard, PRX Quantum (2021)

Beyond the circuit formalism

What cannot be done?

Purifiable processes, reversibility preserving:



M. Araújo, A. Feix, M. Navascués, A. Brukner, Quantum (2017)

Part 4

The cost of a quantum circuit simulation

Results based on:

1. [arXiv:2409.18420](#) [pdf, other] [quant-ph](#)

Exponential separation in quantum query complexity of the quantum switch with respect to simulations with standard quantum circuits

Authors: [Hlér Kristjánsson](#), [Tatsuki Otake](#), [Satoshi Yoshida](#), [Philip Taranto](#), [Jessica Bavaresco](#), [Marco Túlio Quintino](#), [Mio Murao](#)

Abstract: Quantum theory is consistent with a computational model permitting black-box operations to be applied in an indefinite causal order, going beyond the standard circuit model of computation. The quantum switch -- the simplest such example -- has been shown to provide numerous information-processing advantages. Here, we prove that the action of the quantum switch on two n -qubit quantum channels can... [▽ More](#)

Submitted 1 October, 2024; **v1** submitted 26 September, 2024; **originally announced** September 2024.

Comments: 23 pages, 3 figures

2. [arXiv:2409.18202](#) [pdf, other] [quant-ph](#)

Can the quantum switch be deterministically simulated?

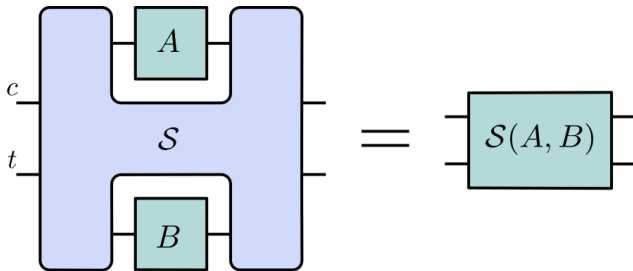
Authors: [Jessica Bavaresco](#), [Satoshi Yoshida](#), [Tatsuki Otake](#), [Hlér Kristjánsson](#), [Philip Taranto](#), [Mio Murao](#), [Marco Túlio Quintino](#)

Abstract: Higher-order transformations that act on a certain number of input quantum channels in an indefinite causal order - such as the quantum switch - cannot be described by standard quantum circuits that use the same number of calls of the input quantum channels. However, the question remains whether they can be simulated, i.e., whether their action on their input channels can be deterministically repr... [▽ More](#)

Submitted 26 September, 2024; **originally announced** September 2024.

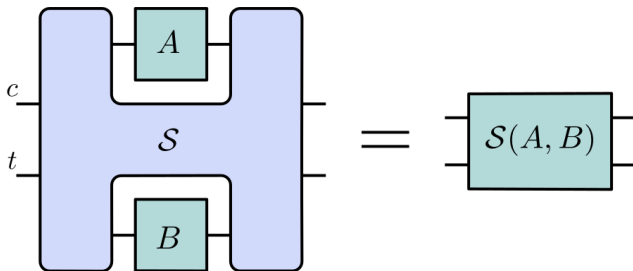
Comments: 16 + 14 pages, 4 + 5 figures

The quantum switch



What can we do with that?

The quantum switch



What can we do with that?

The commuting, anti-commuting game:

Perfect discrimination of no-signalling channels via quantum superposition of causal structures

G. Chiribella, PRA 2012

Witnessing causal nonseparability

M. Araújo, C. Branciard, F. Costa, A. Feix, C. Giarmatzi, Č. Brukner, NJP 2015

Commutation/Anti-Commutation game

(U_A, U_B) is a pair of unitary that

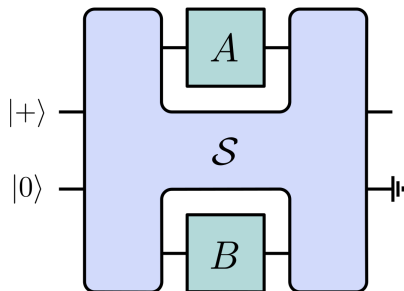
$$U_A U_B = U_B U_A \quad \text{or} \quad U_A U_B = -U_B U_A$$

Commutation/Anti-Commutation game

(U_A, U_B) is a pair of unitary that

$$U_A U_B = U_B U_A \quad \text{or} \quad U_A U_B = -U_B U_A$$

The quantum switch is useful:



Commutation/Anti-Commutation game

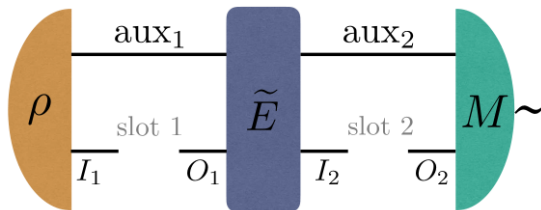
- ▶ Let $\{(U_A^i, U_B^i)\}_{i=1}^N$ be a set of unitaries that commutes or anticommutes.

Commutation/Anti-Commutation game

- ▶ Let $\{(U_A^i, U_B^i)\}_{i=1}^N$ be a set of unitaries that commutes or anticommutes.
- ▶ Given a pair of unitaries at random, can you decide if they commute or anticommute?

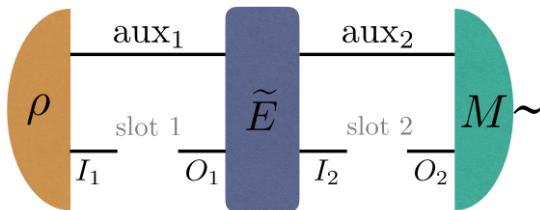
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- ▶ Standard ordered strategy:



Commutation/Anti-Commutation game

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- ▶ Standard ordered strategy:



- ▶ We can find finite sets of unitaries such that $p_{\text{ordered}} \leq 0.87$.

Commutation/Anti-Commutation game

- ▶ Great, the quantum switch is useful!

Commutation/Anti-Commutation game

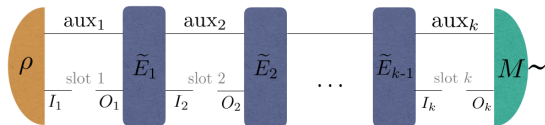
- ▶ Great, the quantum switch is useful!
- ▶ But ...

Commutation/Anti-Commutation game

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- ▶ How big is this advantage? What if we do not have the quantum switch, but we have access to more queries?

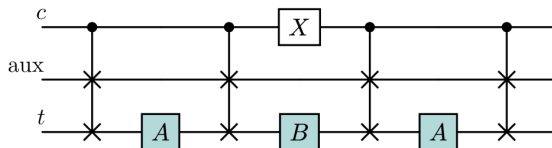
Commutation/Anti-Commutation game

- ▶ Great, the quantum switch is useful!
- ▶ But ...
- ▶ How big is this advantage? What if we do not have the quantum switch, but we have access to more queries?
- ▶ With a single extra query, sequential strategies can decide if (U_A, U_B) commutes or anti-commutes



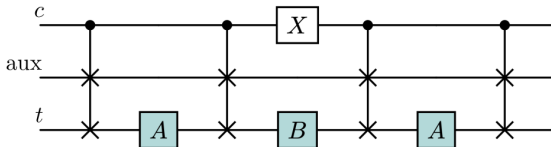
Quantum switch circuit simulation

- Quantum computations without definite causal structure
G. Chiribella, G. M. D'Ariano, P. Perinotti, B. Valiron, PRA (2013)

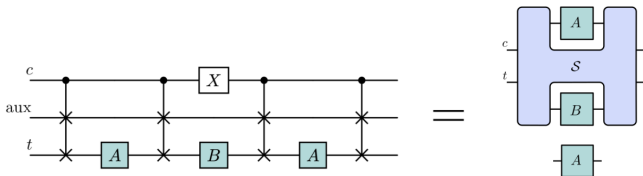


Quantum switch circuit simulation

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G. Chiribella, G. M. D'Ariano, P. Perinotti, B. Valiron, PRA (2013)

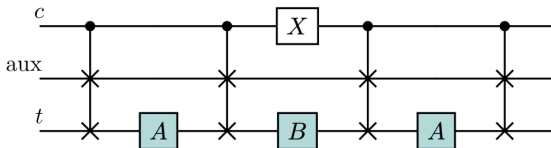


- ▶ If A and B are unitary:

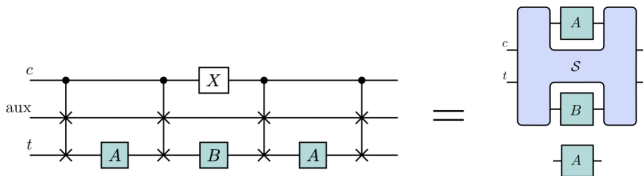


Quantum switch circuit simulation

- ▶ Quantum computations without definite causal structure
G. Chiribella, G. M. D'Ariano, P. Perinotti, B. Valiron, PRA (2013)



- ▶ If A and B are unitary:



- ▶ The switch is essentially useless for query complexity tasks...

Quantum switch circuit simulation

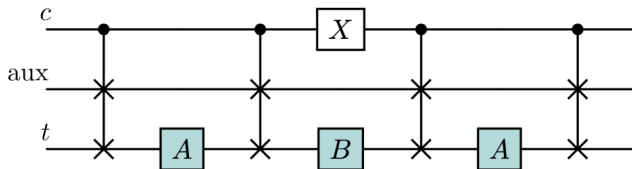
► Wait. . .

Quantum switch circuit simulation

- ▶ Wait. . .
- ▶ What if the operations are not unitary?

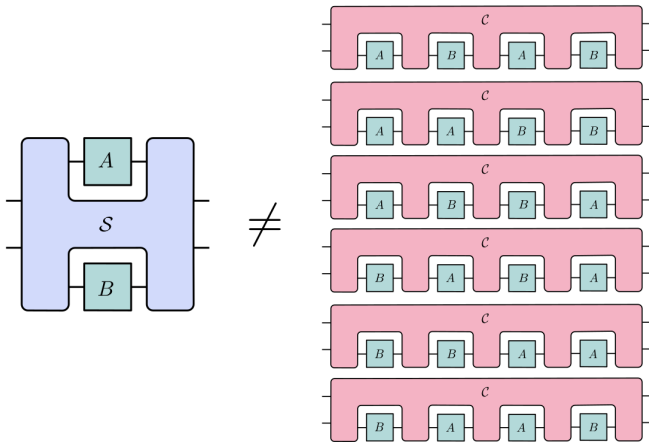
Quantum switch circuit simulation

- ▶ Wait. . .
- ▶ What if the operations are not unitary?
- ▶ E.g., $A(\rho) = B(\rho) = \text{tr}(\rho) \frac{\mathbb{I}}{d}$



Quantum switch circuit simulation

Thm1: There is no quantum circuit that simulates the quantum switch when one extra query of each channel is available.

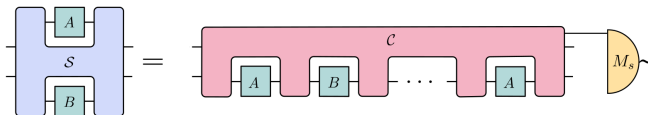


Quantum switch circuit simulation

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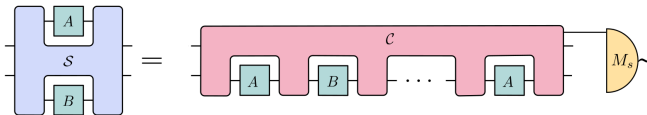
Quantum switch circuit simulation

- ▶ Thm1: There is no quantum circuit that simulates the quantum switch when one extra query of each channel is available.
- ▶ Probabilistic simulation:

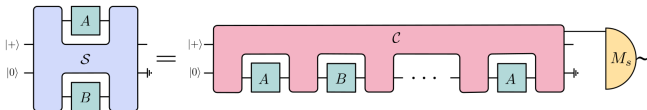


Quantum switch circuit simulation

- ▶ Thm1: There is no quantum circuit that simulates the quantum switch when one extra query of each channel is available.
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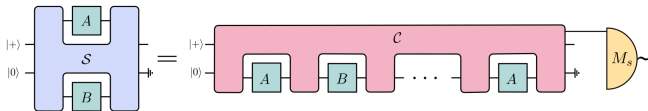


- ▶ Restricted probabilistic simulation:



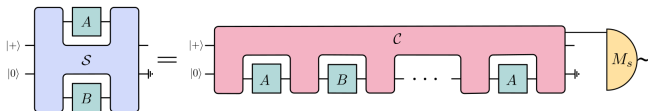
Quantum switch circuit simulation

- How about the probabilities?



Quantum switch circuit simulation

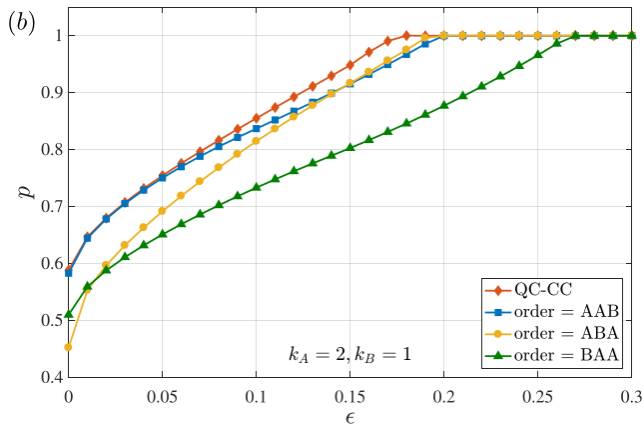
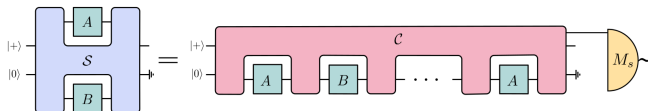
- How about the probabilities?



(k_A, k_B)	order	probability
$(1, 1)$	AB	$p < \frac{4001}{10000}$
$(2, 1)$	AAB	$p < \frac{5715}{10000}$
	ABA	$p < \frac{4919}{10000}$
	BAA	$p < \frac{5001}{10000}$

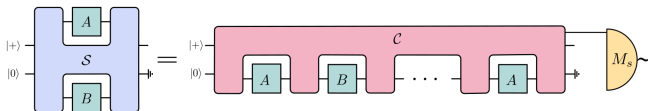
Quantum switch circuit simulation

- The result is also robust, $F(\mathcal{S}, \mathcal{S}_{\text{sim}}) = 1 - \epsilon$



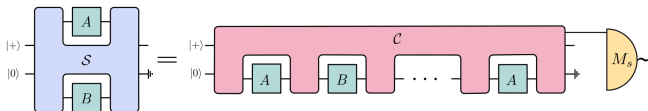
Quantum switch circuit simulation

- How about the probabilities when $k = 4$?



Quantum switch circuit simulation

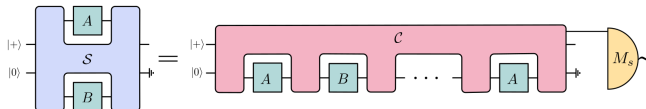
- How about the probabilities when $k = 4$?



(2, 2)	AABB	$p < \frac{8307}{10000}$
	ABAB	$p < \frac{8484}{10000}$
	ABBA	$p < \frac{8695}{10000}$
(3, 1)	AAAB	$p < \frac{8373}{10000}$
	AABA	$p < \frac{6909}{10000}$
	ABAA	$p < \frac{7597}{10000}$
	BAAA	$p < \frac{6845}{10000}$

Quantum switch circuit simulation

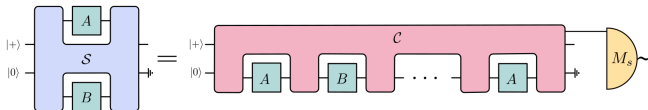
- How about identical channels?



k	order	probability
2	AA	$p < \frac{4001}{10000}$
3	AAA	$p < \frac{6534}{10000}$
4	AAAA	$p = 1 (*)$

Quantum switch circuit simulation

- How about identical channels?



k	order	probability
2	AA	$p < \frac{4001}{10000}$
3	AAA	$p < \frac{6534}{10000}$
4	AAAA	$p = 1 (*)$

If the target is not discarded, $p < 0.942$ for the AAAA order

Quantum switch circuit simulation

► How about unitary channels?

(k_A, k_B)	order (unitary only)	probability
(1, 1)	AB	$p \approx 0.400$
(2, 1)	AAB	$p \approx 0.596$
	ABA	$p = 1$
	BAA	$p \approx 0.607$
(2, 2)	AABB	$p = 1$ (*)
	ABAB	$p = 1$
	ABBA	$p = 1$
(3, 1)	AAAB	$p \approx 0.708$
	AABA	$p = 1$
	ABAA	$p = 1$
	BAAA	$p = 1$ (*)

Quantum switch circuit simulation

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(k_A, k_B)	order (unitary only)	probability
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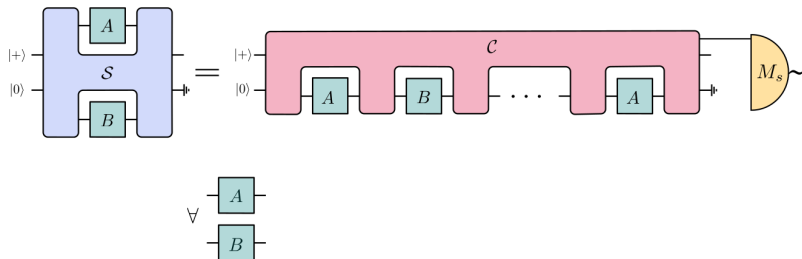
If the target is not discarded, $p < 0.822$ for the AABB order and $p < 0.667$ for the BAAA order

Quantum switch circuit simulation

- ▶ How these results were obtained?

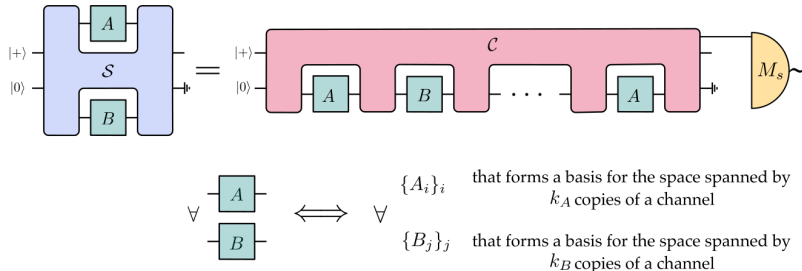
Quantum switch circuit simulation

- ▶ How these results were obtained?
- ▶ Optimise over all inputs:



Quantum switch circuit simulation

- ▶ How these results were obtained?
- ▶ Optimise over finitely inputs:



Quantum switch circuit simulation

► SDP (using splitting conic solver)

given $\{J_i^A\}_i, \{J_j^B\}_j, k_A, k_B$

max p

s.t. $C_s * [(J_i^A)^{\otimes k_A} \otimes (J_j^B)^{\otimes k_B}] = p S * (J_i^A \otimes J_j^B) \quad \forall i, j$

$$C_s \geq 0, \quad C - C_s \geq 0,$$

$$\mathbb{P}(C) = C, \quad \text{tr}(C) = d_{c_I} d_{t_I} d_{A_O}^{k_A} d_{B_O}^{k_B},$$

given $\{J_i^A\}_i, \{J_j^B\}_j, k_A, k_B$

min $\frac{1}{d_{A_I}^{k_A} d_{B_I}^{k_B} d_{c_O} d_{t_O}} \text{tr}(\Gamma)$

s.t. $\sum_{i,j} \text{tr}[R_{ij} (S * (J_i^A \otimes J_j^B))] = 1$

$$\Gamma - \sum_{i,j} R_{ij} \otimes [(J_i^A)^{\otimes k_A} \otimes (J_j^B)^{\otimes k_B}]^T \geq 0$$

$$\Gamma \geq 0, \quad \bar{\mathbb{P}}(\Gamma) = \Gamma,$$

any feasible point that yields some
 $p < 1$
constitutes a valid upper bound

Quantum switch circuit simulation

- ▶ But... is that a mathematical proof?

Quantum switch circuit simulation

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- ▶ No! But, we can extract a proof out of it!

Quantum switch circuit simulation

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- ▶ No! But, we can extract a proof out of it!

Algorithm:

1. Construct symbolic non-floating point operators Γ^{sym} and R_{ij}^{sym} by truncating them and obtaining a symbolic operator with only rational numbers.
2. Force the operators Γ^{sym} and R_{ij}^{sym} to be self-adjoint by making use of the expression $(M + M^\dagger)/2$, which is self-adjoint for any M .
3. Evaluate $t^{\text{sym}} := \sum_{i,j} \text{tr}[R_{ij}^{\text{sym}} (S * (J_i^A \otimes J_j^B))]$, where S, J_i^A , and J_j^B are also symbolic operators. Define $R_{ij}^{\text{ok}} := R_{ij}^{\text{sym}}/t^{\text{sym}}$ for all i, j .
4. Project Γ^{sym} onto the appropriate subspace to obtain $\bar{\mathbb{P}}(\Gamma^{\text{sym}})$.
5. Find $\eta \in \mathbb{R}$ such that $\Gamma^{\text{ok}} := \bar{\mathbb{P}}(\Gamma^{\text{sym}}) + \eta \mathbb{1} \geq 0$ and $\Gamma^{\text{ok}} - \sum_{i,j} R_{ij}^{\text{ok}} \otimes (J_i^A \otimes J_j^B)^T \geq 0$.
6. Output the quantity $\text{tr}(\Gamma^{\text{ok}})/d_{c_I} d_{t_I} d_{A_O}^{k_A} d_{B_O}^{k_B}$, which is a rigorous upper bound of the primal problem.

Quantum switch circuit simulation

- ▶ Very nice... But with the computer we are limited to a few queries...

Quantum switch circuit simulation

- ▶ Very nice. . . But with the computer we are limited to a few queries. . .
- ▶ Thm2: Let $d = 2^n$ be the dimension of the target state. If $k_A = 1$ and $k_B < 2^n$, there is no quantum circuit simulation of the quantum switch.

Quantum switch circuit simulation

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We then analyse the case of Pauli operations using a differentiation technique (from Analytical lower bound on query complexity for transformations of unknown unitary operations, T. Otake, S. Yoshida, M. Murao).

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This allows us to note that imposing the simulation to hold has very strong implications.
In particular, its eigendecomposition cannot be compatible with QCQC processes.

Quantum switch circuit simulation

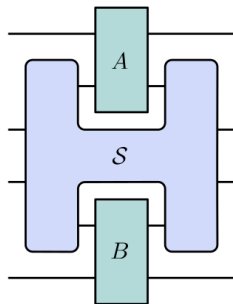
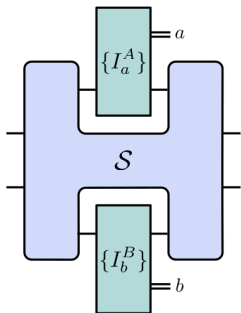
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Quantum switch circuit simulation

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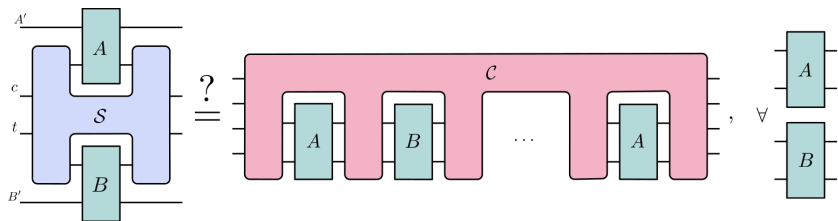
Quantum switch circuit simulation

- ▶ How to go beyond?
- ▶ The quantum switch is not restricted to single-partite channels.
- ▶ How about instruments, bipartite channels?



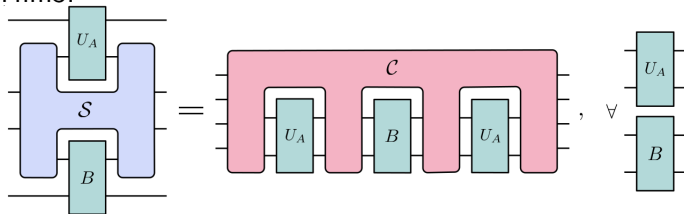
Quantum switch circuit simulation

General simulation:



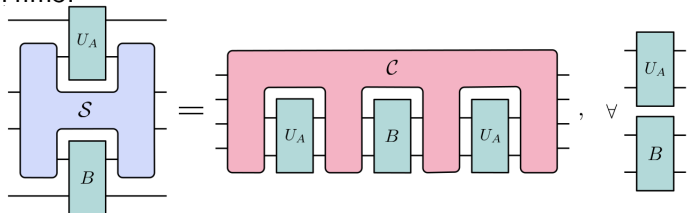
Quantum switch circuit simulation

► Thm3:

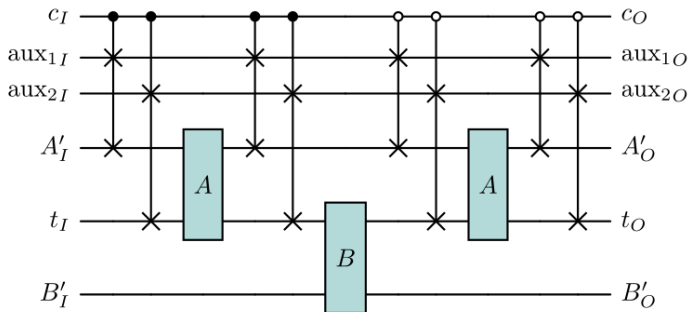


Quantum switch circuit simulation

► Thm3:



► The circuit:



Outlook

- ▶ HOQOs are a powerful tool to describe and analyse quantum operations (SDP, group representation, etc)

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- ▶ HOQOs are a powerful tool to describe and analyse quantum operations (SDP, group representation, etc)
- ▶ Channel transformations, channel measurement, channel comparison, oracle based tasks
- ▶ What can we say about “gray box” scenarios?
- ▶ HOQO methods and approach to adjacent fields?

Thank you

